

Part 1: Your Hypotheses

Before running any experiments, predict the rejection rate for each defense configuration against each attack type. Write a percentage (0% or 100%) in each cell.

Experiment	Immediate Replay	Delayed Replay	Modified Replay
No defense	0%	0%	0%
Timestamp only	0%	100%	0%
Counter only	100%	100%	0%
All defenses	100%	100%	100%

Part 2: Experiment Results

Experiment 1: No Defense (Baseline)

Attack Type	Accepted	Rejected	Rejection Rate
Immediate replay	5	0	0%
Delayed replay	5	0	0%
Modified replay	5	0	0%

Experiment 2: Timestamp Only

Attack Type	Accepted	Rejected	Rejection Rate
Immediate replay	5	0	0%
Delayed replay	0	5	100%
Modified replay	5	0	0%

Experiment 3: Counter Only

Attack Type	Accepted	Rejected	Rejection Rate
Immediate replay	0	5	100%
Delayed replay	0	5	100%
Modified replay	5	0	0%

Experiment 4: All Defenses

Attack Type	Accepted	Rejected	Rejection Rate
Immediate replay	0	5	100%
Delayed replay	0	5	100%
Modified replay	0	5	100%

Part 3: Summary Table

Compile all your results into a single table:

Experiment	Immediate Replay	Delayed Replay	Modified Replay
No defense	0%	0%	0%
Timestamp only	0%	100%	0%
Counter only	100%	100%	0%
All defenses	100%	100%	100%

Part 4: Screenshots

Terminal output (experiment summary):

```
[RESULTS]
Messages tested: 5
Accepted: 5
Rejected: 0
Rejection rate: 0%

=====
EXPERIMENT SUMMARY
=====
Defense      Attack      Accepted  Rejected  Rate
-----
none         immediate    5          0         0%
none         delayed      5          0         0%
none         modified     5          0         0%

[RESULTS]
Messages tested: 5
Accepted: 5
Rejected: 0
Rejection rate: 0%

=====
EXPERIMENT SUMMARY
=====
Defense      Attack      Accepted  Rejected  Rate
-----
counter      immediate    0          5        100%
counter      delayed      0          5        100%
counter      modified     5          0         0%

[SAVED] Results saved to experiment_results.json
```

```
[RESULTS]
Messages tested: 5
Accepted: 5
Rejected: 0
Rejection rate: 0%

=====
EXPERIMENT SUMMARY
=====
Defense      Attack      Accepted  Rejected  Rate
-----
timestamp    immediate    5          0         0%
timestamp    delayed      0          5        100%
timestamp    modified     5          0         0%

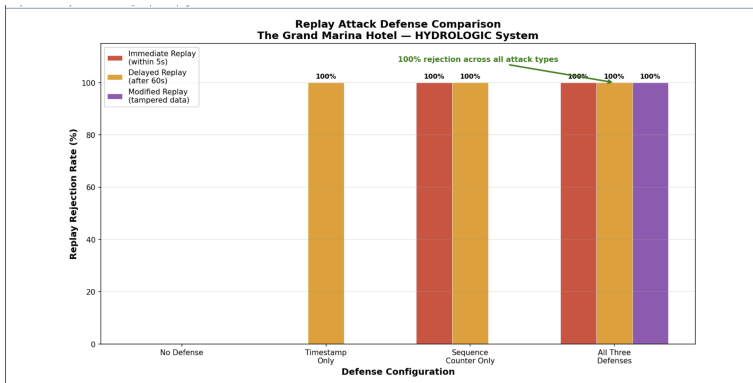
[SAVED] Results saved to experiment_results.json
```

```
[RESULTS]
Messages tested: 5
Accepted: 0
Rejected: 5
Rejection rate: 100%

=====
EXPERIMENT SUMMARY
=====
Defense      Attack      Accepted  Rejected  Rate
-----
all          immediate    0          5        100%
all          delayed      0          5        100%
all          modified     0          5        100%

[SAVED] Results saved to experiment_results.json
```

defense_comparison.png chart:



Part 5: Hypothesis Comparison

1. Which predictions were correct?

All my predictions were correct

2. Which results surprised you? Were there any attack/defense combinations you expected to block that didn't, or vice versa?

No Result surprised me, it was pretty obvious.

3. Why did individual defenses have gaps? Think about what each defense checks and what it doesn't.

Because of the nature of their validations.